

Premises: 1) $\forall x (W(x) \Rightarrow S(x) \vee PS(x))$ 2. $W(\text{Monday}) \vee W(\text{Friday})$
 3) $\neg S(\text{Tuesday})$ 4) $\neg PS(\text{Friday})$

Steps	Reason
1. $\forall x (W(x) \Rightarrow S(x) \vee PS(x))$	Premise
2. $W(\text{Monday}) \vee W(\text{Friday})$	Premise
3. $\neg S(\text{Tuesday})$	Premise
4. $\neg PS(\text{Friday})$	Premise
5. $W(\text{Monday}) \Rightarrow S(\text{Monday}) \vee PS(\text{Monday})$	Universal instantiation on 1
6. $\neg W(\text{Monday}) \vee S(\text{Monday}) \vee PS(\text{Monday})$	Conditional disjunction equivalence on 5
7. $W(\text{Friday}) \vee S(\text{Monday}) \vee PS(\text{Monday})$	Resolution on 2, 6
8. $W(\text{Friday}) \Rightarrow S(\text{Friday}) \vee PS(\text{Friday})$	Universal instantiation on 1
9. $\neg W(\text{Friday}) \vee S(\text{Friday}) \vee PS(\text{Friday})$	Conditional disjunction equivalence on 8
10. $S(\text{Monday}) \vee PS(\text{Monday}) \vee S(\text{Friday}) \vee PS(\text{Friday})$	Resolution on 7, 9
11. $S(\text{Monday}) \vee PS(\text{Monday}) \vee S(\text{Friday})$	Disjunctive syllogism on 10, A
12. $W(\text{Tuesday}) \Rightarrow S(\text{Tuesday}) \vee PS(\text{Tuesday})$	Universal instantiation on 1
13. $\neg W(\text{Tuesday}) \vee S(\text{Tuesday}) \vee PS(\text{Tuesday})$	Conditional disjunction equivalence on 12
14. $(S(\text{Tuesday}) \vee PS(\text{Tuesday})) \vee \neg W(\text{Tuesday})$	Commutative law on 13
15. $S(\text{Tuesday}) \vee (PS(\text{Tuesday}) \vee \neg W(\text{Tuesday}))$	Associative law on 14
16. $PS(\text{Tuesday}) \vee \neg W(\text{Tuesday})$	Disjunctive syllogism on 15, 3